

4(a)	straight line through origin shows that a is proportional to x	B1
	negative gradient shows that a is in opposite direction to x	B1
4(b)(i)	$a_0 = \omega^2 x_0$ or $a = -\omega^2 x$ or $\omega^2 = -\text{gradient}$	C1
	$\omega = \sqrt{(0.40 / 0.050)}$ $= 2.8 \text{ rad s}^{-1}$	A1
4(b)(ii)	$k = \omega^2 L$ $= 2.8^2 \times 1.24$	C1
	$= 9.7 \text{ m s}^{-2}$	A1
4(c)	(increasing L causes) ω to decrease or energy ($= \frac{1}{2} m \omega^2 x_0^2$) $= \frac{1}{2} m k x_0^2 / L$ (and L increases)	M1
	so amplitude increases	A1